AEGIS FORECASTING IMPROVEMENT · TESSERACT V3

Project Documentation

Version V3 · Stage 1 baseline · Build 2026-06-25
Derived (read-only) from the governed Markdown source aegis_v3_project_documentation.md

Contents

AEGIS Forecasting Improvement / Tesseract V3 — Project Documentation

Version: V3 Status: Active baseline (Stage 1 — Methodology Documentation) Date: 2026-06-25 Scope of this document: formal project write-up for the V3 dashboard and its read-only data contract. This document describes the approach; it does not change any data artifact, forecast, interval, champion decision, metric, or governance outcome.

---

1. What is AEGIS Forecasting Improvement / Tesseract V3

AEGIS Forecasting Improvement (internally "Tesseract V3") is a governed forecasting review platform for enterprise HDD-region capacity series. It takes forecasting evidence produced by an upstream pipeline and presents it through a read-only Shiny dashboard so that reviewers can inspect forecasts, model comparisons, intervals, capacity views, and governance decisions in one place.

V3 is the final version line. It is built as a controlled clone of V2 (the functional MVP baseline) and evolves it with documentation, an AI explanation layer, a more robust forecasting model, and an automated daily refresh — each introduced as a separate, governed stage.

2. What problem it solves

Forecasting outputs and their governance evidence are normally scattered across SQL extracts, processing scripts, model-lab notebooks, tournament results, and audit documents. Reviewers cannot easily answer: which model won, how accurate it is, how wide the uncertainty is, what the capacity runway looks like, and whether the decision was governed. V3 consolidates that evidence into a single, consistent, read-only view backed by governed artifacts, so review and sign-off are auditable and repeatable.

3. Central architectural rule: Shiny does not cook data

The single most important rule of the platform:

> The Shiny dashboard is a **read-only consumer**. It never downloads from Tesseract, never cleans data, never trains or recalculates models, never recomputes forecasts or intervals, and never writes or edits artifacts.

All data is produced and governed by the **upstream pipeline**. The dashboard loads the governed CSVs once at startup through a read-only loader and renders them as-is. If a number is wrong upstream, it is fixed upstream — not in the app.

4. General architecture

The system has two clearly separated halves (see `docs/architecture/aegis_v3_architecture_diagram.mmd` / `.png`):

- **Upstream pipeline (producer):** Tesseract v2 (SQL) → ingestion (Python) → `data/raw/` → processing/validation → `data/processed/` (governed CSV contract) → Model Lab (training, backtest, tournament, champion selection) → forecast, interval, and governance/reference artifacts.
- **Shiny dashboard (consumer):** a read-only data loader feeds the Forecasting, Models, Governance, and Reference page groups. No write-back path exists.

Planned/optional V3 components (a daily refresh orchestrator and an AI/LLM explanation layer) are marked as planned and are not yet implemented.

5. Data sources

- **Primary source:** `TesseractEarthDW.dbo.forecast_substrateBE_hdd_region` (Scenario = Enterprise, ValueType = Forecast-Mean), queried by the Python ingestion layer and exported to `data/raw/`.
- **Governed processed contract (`data/processed/`):** `forecasts.csv`, `actuals.csv`, `entities.csv`, `run_metadata.csv` — validated and reshaped with no shifted dates, no imputations, and no recompute.

6. Principal artifacts

- **Forecast/backtest:** `forecasts.csv`, `actuals.csv`, and `forecast_viewer_model_outputs.csv` (per-model backtest used by Viewer/Accuracy).
- **Intervals:** calibrated 80% prediction bands (see Section 8).
- **Model Lab closure pack:** key results, model universe, champion summary, risk register, next steps.
- **Tournament:** preliminary standings, model scorecard, pairwise evidence.
- **Governance:** audit findings, sanity review, champion conditions, and the approved dashboard language.

The Reference → Artifacts page lists the full governed registry and its availability; the dashboard never edits these files.

7. Forecasting

The Forecasting page group presents series-level evidence:

- **Viewer:** actuals vs forecast per series, with model overlays from the backtest outputs.
- **Accuracy:** per-model backtest error diagnostics.
- **Forecast:** the governed forward forecast per series.
- **TTL / Capacity View:** time-to-limit / capacity runway (see Section 9).

All values are read directly from governed artifacts.

8. Intervals — 80% calibrated up to 60 days

The dashboard shows **80% prediction intervals that are empirically calibrated up to a 60-day horizon**. Beyond the calibrated horizon, bands are not asserted as calibrated. Intervals are produced upstream by the Model Lab calibration step; the dashboard only displays them and does not recompute coverage or widen/narrow bands.

9. TTL / Capacity View

The TTL / Capacity View estimates a capacity runway (time-to-limit) from the governed forecast. **Supply / capacity inputs remain prototype / simulated** unless and until a validated supply artifact exists. Where supply is simulated, the view is a planning prototype, not a production capacity commitment, and is labeled accordingly. This view is read-only like every other page.

10. Models / Tournament / Champion

- **Universe:** the full set of candidate models considered.
- **Tournament:** preliminary standings, scorecard, and pairwise evidence used to rank models.
- **Champion:** the governed selected model and the conditions attached to its selection.

The champion is a **governed decision**. The dashboard displays the current governed champion; it does not select, change, or auto-promote any model.

11. Governance / Risks / Audit

- **Risks:** the governed risk register and deferred-model / conditional context.
- **Audit:** the audit trail (audit findings, sanity review, verification verdict).

These reflect upstream governance decisions verbatim. The dashboard is evidence of governance, not an actor in it.

12. Reference / Version / Freshness

- **Artifacts:** the governed artifact registry and downloads.
- **Methodology:** this approach, the data pipeline, and the architecture diagram.
- **Version:** build/runtime metadata, app version, policy, champion line, and the data snapshot freshness (forecast version, run date, coverage).

Freshness reflects the build date of the governed data contract, not a live query.

13. Current limitations

- The dashboard is strictly read-only and reflects the last governed build; it is not live against Tesseract.
- Intervals are only asserted as calibrated up to 60 days.
- TTL / Capacity supply inputs are prototype / simulated unless a validated supply artifact exists.
- The deep-learning challenger (FastNeuralAR_MLP) underperforms and is a candidate for replacement (planned, not yet done).

- No automated daily refresh is in place yet; updates are produced by running the upstream pipeline.
- No AI/LLM explanation layer is wired yet.

14. Roadmap (V3)

- **Stage 0 (done):** controlled clone of V2 → V3, parity validated, version bumped.
- **Stage 1 (this stage):** methodology documentation + architecture diagram.
- **Stage 2 (planned):** AI explanation layer with a provider abstraction (none / mock / azure_openai / local), starting with mock/static summaries; preference for Azure OpenAI or an approved corporate model; the LLM only explains artifacts and never computes.
- **Stage 3 (planned):** evaluate and replace the FastNeuralAR_MLP model against the engine and existing backtest (neural vs gradient-boosting), with no auto-promotion.
- **Stage 4 (planned):** daily refresh orchestrator (~10:00) — local benchmark first to measure end-to-end duration, then a robust Azure / approved internal scheduler.

15. How to interpret the dashboard

- Treat every number as **as-of the last governed build**, not live.
- Use Viewer/Accuracy to judge model behavior, Forecast for the forward path, and intervals for uncertainty (only up to 60 days).
- Use Tournament/Champion to understand model selection, and Governance for the decision trail.
- Use Reference → Version to confirm which data snapshot you are looking at.

16. What must NOT be assumed as final production

- TTL / Capacity supply figures where supply is simulated/prototype.
- Interval coverage beyond the 60-day calibrated horizon.
- The current deep-learning challenger as a final model choice.
- Any planned/optional component (daily refresh, AI/LLM layer) as already operating.
- Champion changes from any automated process — champion changes require human governance approval.